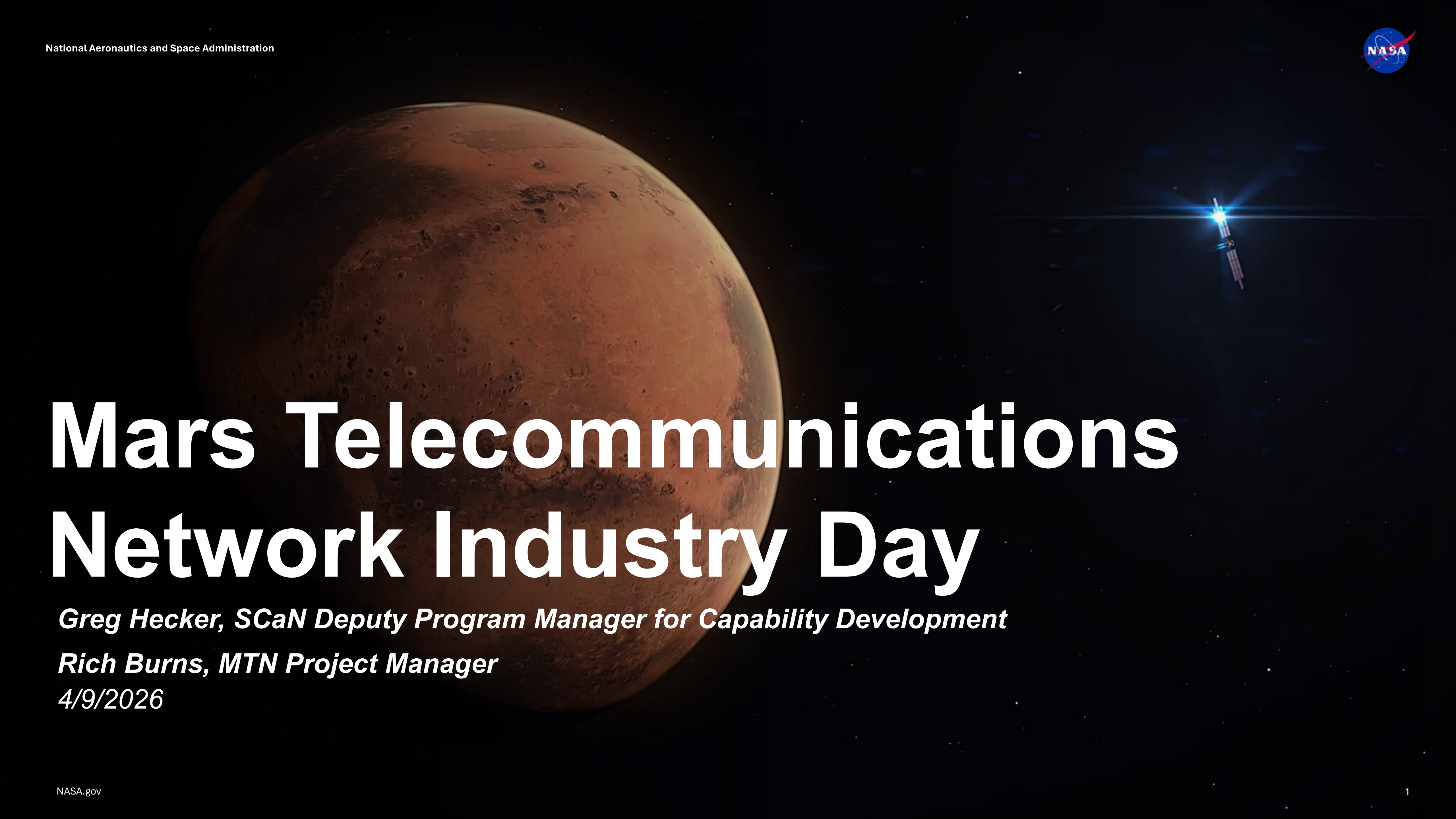
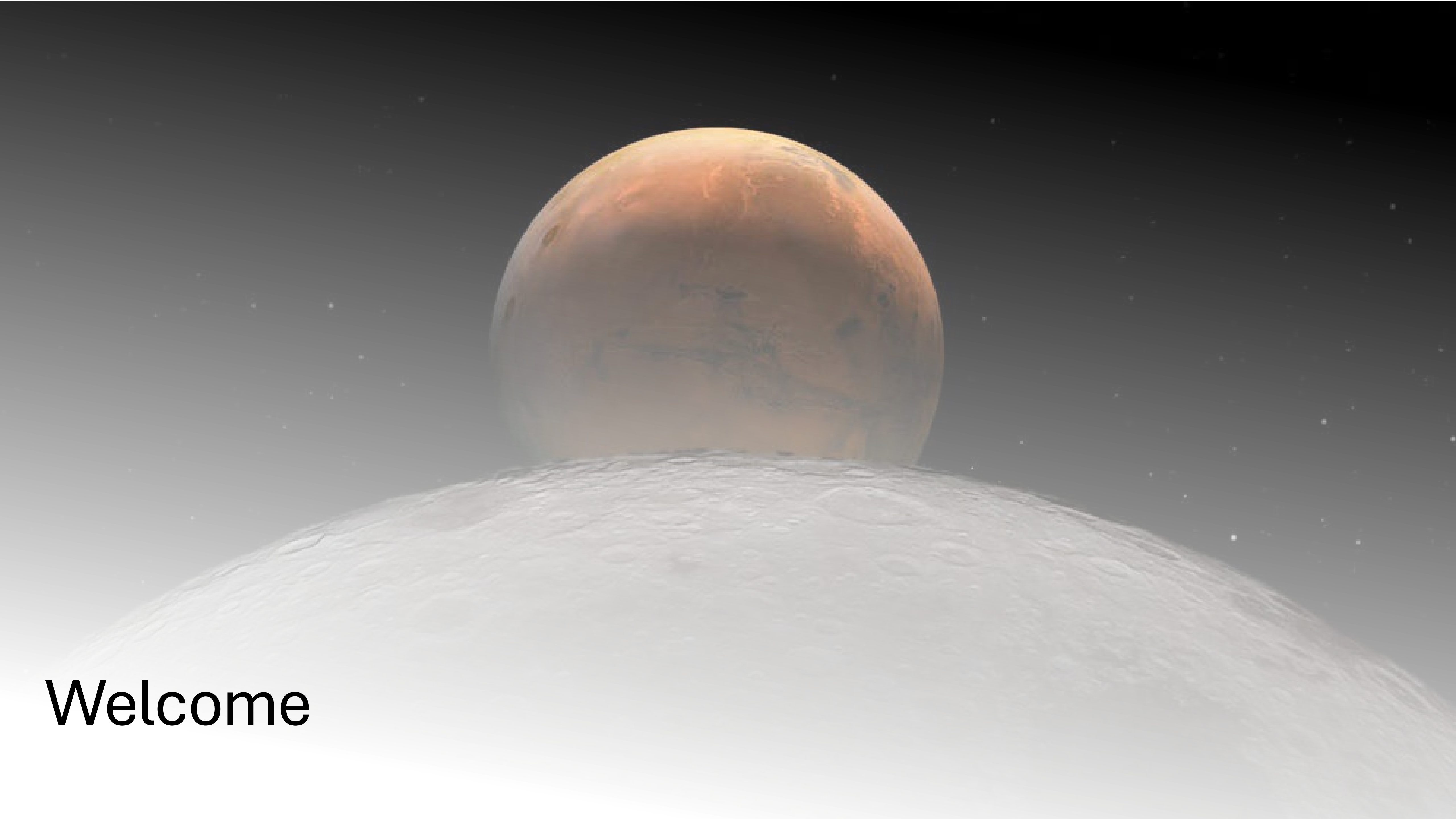


Mars Telecommunications Network Industry Day

Greg Hecker, SCaN Deputy Program Manager for Capability Development

Rich Burns, MTN Project Manager

4/9/2026



Welcome

Logistics

Agenda

Time	Topic	Presenter	Location
8:30 – 8:40 am	Welcome	Cynthia Simmons, Goddard Space Flight Center Director	B3 Auditorium
8:40 – 8:50 am	Logistics	Kelly Hyde, Project Implementation Manager, MTN	B3 Auditorium
8:50 – 9:05 am	Challenges & Opportunities	Greg Heckler, Deputy Program Manager for Capability Development	B3 Auditorium
9:05 – 10:00 am	MTN Project	Rich Burns, Project Manager, MTN	B3 Auditorium
10:00 – 10:30 am		Break	
10:30 am – 11:30 am	Q&A	Procurement	B3 Auditorium

MTN Industry Day Objectives and Disclaimer

Objectives:

- Provide a forum to further explain the MTN objectives, requirements, and procurement structure.
- Provide the opportunity for NASA to hear from industry on the dFRP.
- Advance toward a mature, thoroughly considered RFP that sets the stage for MTN success.

Notes:

- All information provided during Industry Day is preliminary.
- In the event of a discrepancy between the information you receive today and the information in the final RFP, the final RFP is the controlling document.
- ***The final RFP will take precedence over the dRFP or any information you receive today.***

Logistics

All conversations in this session are public

- No Controlled Unclassified Information (CUI) or Proprietary Information should be shared.

No photographs or recording

- Industry day presentation will be posted on sam.gov

Wi-Fi access:

- Instructional sheets are available where you checked in.

Restrooms:

- Exit auditorium, turn left, continue down the hallway, up the small staircase; restrooms located on the right.

Reminder:

- Comments/Responses for the Draft RFP are due 4/13/26 per the Cover Letter

Question & Answer Instructions

If you have a question, please submit it to the QR code.

Procurement will take 30 minutes after the briefing to examine questions and prepare our responses.

After the break, we will reconvene, and address questions submitted.

Questions may not be answered today, but questions and answers will be posted on SAM.gov.

Questions posed in 1:1 sessions are unlikely to receive a response today.

Point of Contact

All communications regarding this dRFP should go through Alisha Mercadante (alisha.m.mercadante@nasa.gov), Carlos McKenzie (Carlos.R.Mckenzie@nasa.gov), Antwan Reid at (Antwan.G.Reid@nasa.gov), as listed on SAM.gov.

Challenges & Opportunities

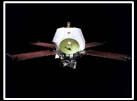
MARS TELECOMMUNICATIONS NETWORK

Mars Exploration

1964 | 1971 | 1975 | 1996 | 2011

Early Reconnaissance

Mariner 4 (1964)



Mariner 9 (1971)

Mariner 6 and 7 (1969)

Search for Life

Viking 1 and 2 (1975)

Follow the Water

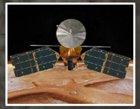
Mars Global Surveyor (1996)

Mars Pathfinder and Sojourner (1996)

Mars Odyssey* (2001)

Mars Express* (2003)

Spirit and Opportunity (2003)



MRO* (2005)

Phoenix (2007)

MSL Curiosity (2011)

Expanded Science

MAVEN* (2013)

ExoMars TGO* (2016)

InSight and MarCO-A/B (2018)

EMM Hope (2020)

Mars 2020 Perseverance and Ingenuity (2020)

* = MRN

Mars and Deep Space Regimes

SCaN initiated Mars and Deep Space strategic planning, building on its success partnering with industry in near Earth and cislunar space

Earth ⟶ Cislunar ⟶ Mars and Deep Space

1

Adopt commercial solutions to provide global, low-latency, high-throughput services to users

Communications Services Project well underway with seven industry partners for near Earth space relay

Near Space Network is validating new DTE services and partners

2

Leverage commercial and international partners to deliver cislunar C&PNT capacity and capability

Lunar relay services partner Intuitive Machines progressing in development and deployment

Lunar Exploration Ground Segment commercial services validation underway

3

Develop a cohesive approach across the agency to identify and close projected gaps

Complete remaining DSN antenna builds by 2030

Integrate future mission needs from across the Agency

Engage in market research

Implement “MTO” requirement found in P.L. 119-21

New capability is needed to meet future agency needs

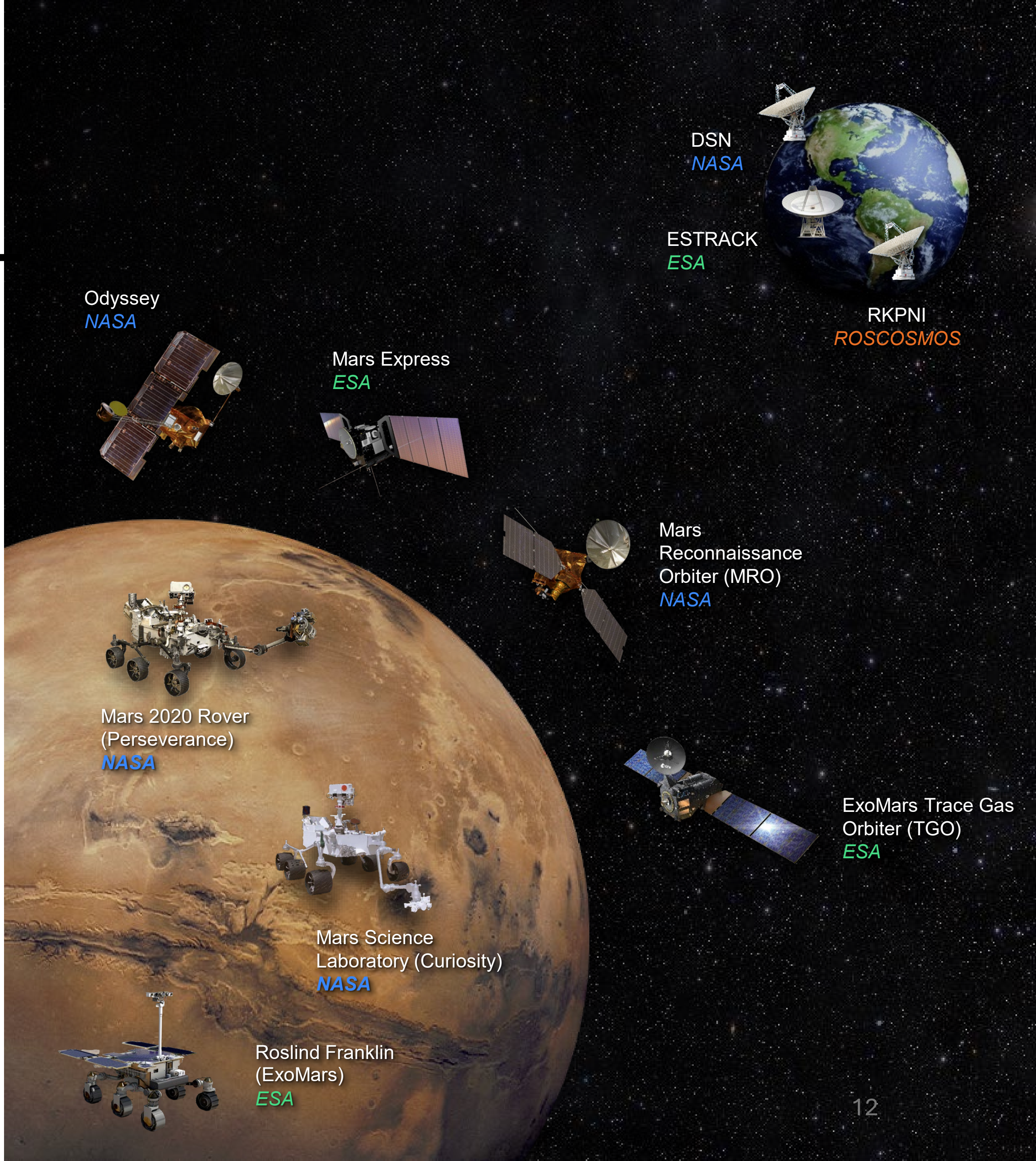
Today's Mars Relay Network (MRN) is operated by the Science Mission Directorate's Mars Exploration Program

Data is delivered through a network of hybrid science/relay orbiters and ground stations, including the Deep Space Network and ESTRACK

However, as these orbiters age and approach retirement, the system's capabilities will steadily decline.

The challenge for MRN's successor:

- **Capitalize on the opportunity enabled by P.L. 119-21**
- **Maintain continuity of MRN services by achieving a 2028 launch**
- **Deploy dedicated infrastructure and new capabilities to advance the long-term Mars C&PNT architecture**



MTN Project

Budget Opportunity

Working Families Tax Cut Act (P.L. 119-21) Provided $700M for Mars Relay:

- Robust continuous comm for Mars Sample Return (MSR) and future Mars surface, orbital, and human exploration missions
- Supporting autonomous operations, onboard processing, and extended mission duration capabilities
- Selected from among commercial proposers for MSR
- Obligation of funds by end of Fiscal Year (FY) 2026; delivery by end of 2028

FY26 President's Budget Request (PBR) Included Tens of Millions for:

- Improved relay infrastructure between Mars and Earth
- Prioritizing partnerships with commercial partners

MTN is proceeding as a single project:

- Developing a single Agency approach for all users
- Determining one set of requirements and life cycle cost

NASA Organization

Exploration Systems Development Mission Directorate (ESDMD)

- Funding organization and primary stakeholder

Space Communications and Navigation (SCaN)

- Program Office

Goddard Space Flight Center

- Project Office, Engineering, Safety & Mission Assurance, Procurement

Glenn Research Center

- Engineering, Telecommunications

Jet Propulsion Laboratory

- Engineering, Mars / Mars Relay Network subject matter expertise

Scope & Architecture Needs

Scope

- Initiate procurement of an initial Mars communications relay mission to support enabling capability and risk reduction for future Moon to Mars (M2M) Architecture.
  - Given this direction and its timeframe, use a tailored approach to integrate the pre-formulation across Agency teams.
  - Primary goal: Address the transition from current Mars science assets to a suitable alternative.

Architecture Needs

- Partially fulfill M2M Architecture use cases UC-C-103 M, UC-C-104 M, UC-C-105 M, UC-C-202 M
- Provide support to existing operational and planned capabilities for Mars surface science missions through at least the mid-2030s
- Provide support to human-scale EDL demonstration missions
- Provide support to technology demonstrations or assets associated with either of the above
- Identify and implement key features to ensure compatibility with future human mission relay assets

Functions		Use Cases	
Provide communications and data exchange between assets in deep space and/or Mars vicinity and Earth	FN-C-104 M	Communications and data exchange between assets in deep space and/or Mars vicinity and Earth	UC-C-104 M
Provide communications and data exchange between assets in deep space and/or Mars vicinity	FN-C-107 M	Communications and data exchange between assets in deep space and/or Mars vicinity	UC-C-105 M
Provide communications and data exchange between assets in deep space and/or Mars vicinity and the Martian surface	FN-C-103 M	Communications and data exchange between assets in deep space and/or Mars vicinity and the Martian surface	UC-C-103 M
Provide position, navigation, and timing data on the Martian surface	FN-C-201 M	Determination of positioning, navigation, and timing for crew and assets at exploration sites on the Martian surface	UC-C-202 M

ESDMD Management Directive (ESDMD-MD-006)

MTN Objectives

Strategic Objectives for the Mars Relay have been derived in accordance with the August 2025 ESDMD/SAO memorandum and updated as a result of the Mission Concept Study and MTN project analysis and stakeholder feedback:

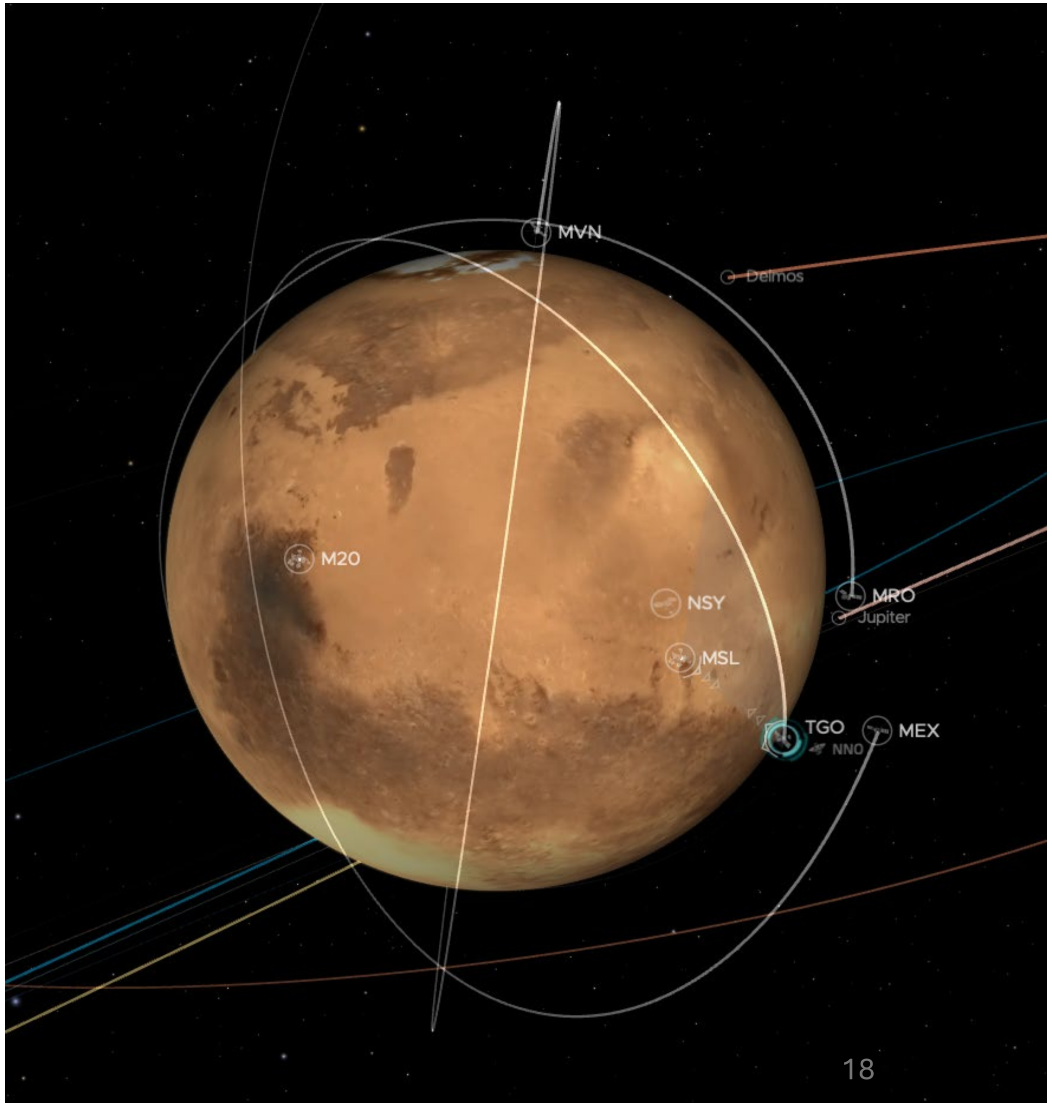
#	Objective:
1	Provide communication and data exchange between assets in the Mars vicinity, the Mars surface, and Earth anticipated to operate at Mars through 2035.
2	Provide Doppler, range, and time transfer to support positioning, navigation, and timing for assets anticipated to operate at Mars through 2035.
3	Provide communication services to existing operational missions.
4	Provide communications, Doppler, range, and time transfer services to the Entry, Descent, and Landing (EDL) demonstration missions anticipated to operate at Mars through 2035.
5*	Provide spacecraft bus integration, accommodation, and communication services to a science payload provided by NASA. *included in pending update to dRFP

Schedule Drivers

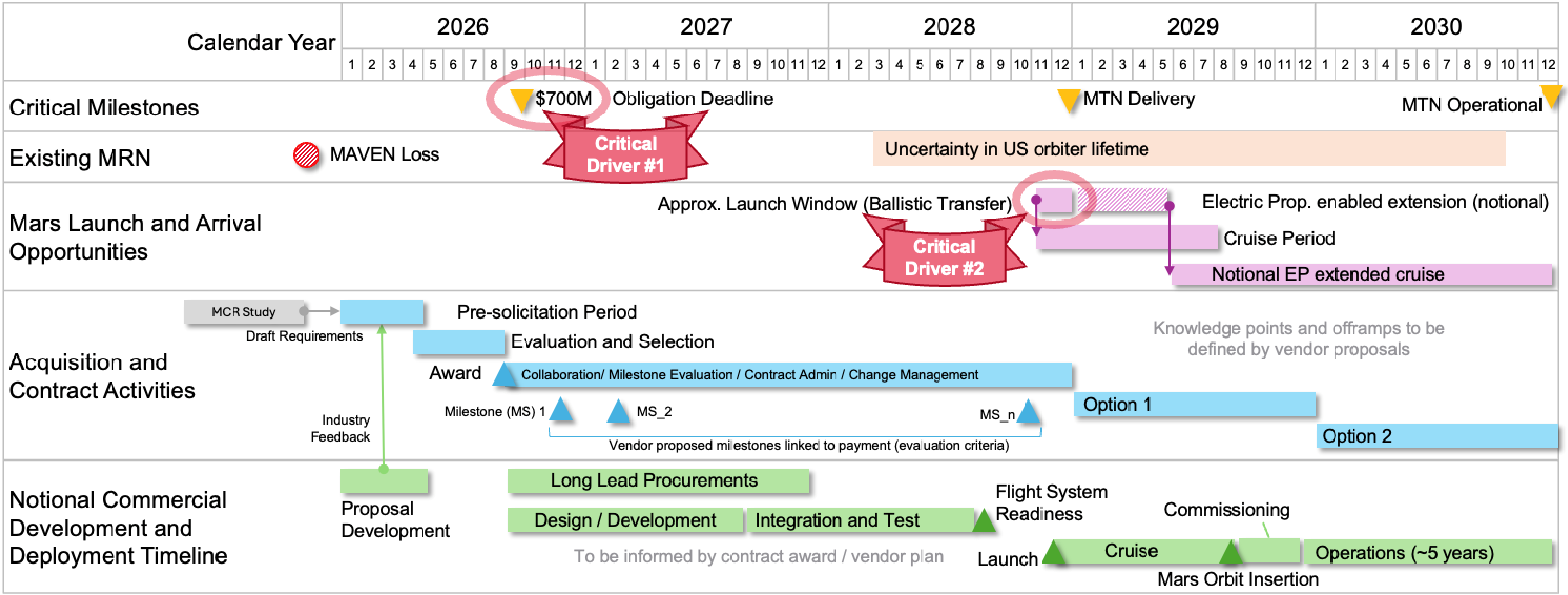
MTN is a schedule-driven project

- P.L. 119-21 requires obligation of funding by September 30, 2026
  - Drives procurement schedule
- P.L. 119-21 requires delivery to the Administration by December 31, 2028
  - Statutory necessity
- Lack of Mars Relay Network robustness
  - Drives operational readiness on fastest implementable timeline
  - MTN-REQ-013 – MTN shall achieve full operational readiness no later than end of calendar year 2030

Procurement and development schedules are paramount.



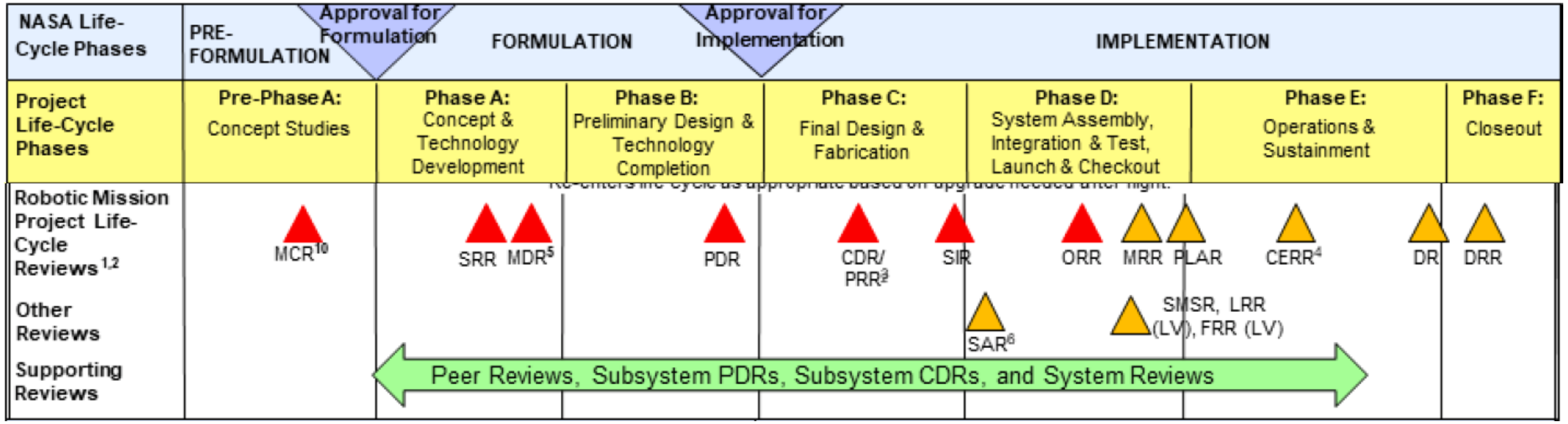
MTN Lifecycle



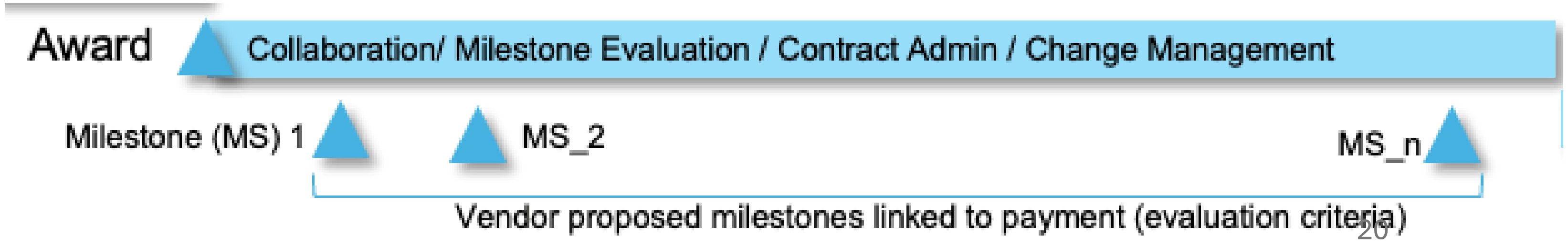
Schedule is primary challenge: <9 months acquisition, 22 months flight systems development/test
Leverage vendor processes, schedule, milestones (maturity of which are evaluation criteria)

MTN Lifecycle

- MTN provides NASA standard practices / review points of 7120.5 and associated milestones for reference
- Each offeror identifies their own milestones/deliverables aligned with corporate practices
- NASA evaluates the proposed milestones and associated payment schedule for reasonableness and effectiveness:
  - Schedule realism
  - Technical depth and content maturity (hardware / results-oriented preferred to reviews/reporting)
  - Payments weighted toward operational readiness demonstrate partner commitment
  - Partner-defined incentives:
    - Readiness to ship
    - Mars Relay commissioning
    - Annual ops requirements met



MTN milestones (to be proposed)



Nominal Procurement Schedule

- Multi-step process described in FAR 15.101(e)
- Potential offerors highly encouraged to submit nonbinding information addressing their statutory eligibility
- The Government will evaluate each potential offerors' submission against the criteria identified OBBBA and provide the following in response to potential offerors:
- Potential offerors considered:
  - likely to be viable competitors will be encouraged to participate in the resultant acquisition.
  - unlikely to be viable competitors will be advised accordingly with the general basis for that opinion.
- Potential offerors, regardless of the advisory result, may still choose to participate in the resultant acquisition.

#	Milestone / Phase	Date	Status
1	Mission Concept Review (MCR)	Week of 1/19	Completed
2	MCR Draft Report	26-Jan	Completed
3	Draft Acquisition Strategy Brief	2-Feb	Completed
4	Pre-solicitation Notice Released	23-Feb	Completed
5	Project Approved to Proceed	16-Feb	Completed
6	Final Acquisition Strategy Brief	23-Feb	Completed
7	Project Lifecycle Defined	2-Mar	Completed
8	Requirements Finalized	9-Mar	Completed
9	Draft Solicitation	2-Apr	Completed
10	Staturory eligibility inputs due	7-Apr	Completed
11	Industry Day	9-Apr	
12	Industry Responses to dRFP	13-Apr	
13	NASA Advisory Feedback to Industry	15-Apr	To be confirmed
14	Final Solicitation	1-May	
15	Written Proposals Due	1-Jun	
16	Proposals and Orals Completed	15-Jun	

Life Cycle Alignment with CLINs

Design/Implementation Phases

Preliminary Design → Detailed Design → Fabrication, Build, Assembly &Test → Launch Vehicle Integration → Launch & Post-Launch → Operations

Contract Line Item Number (CLIN)

CLIN 1 – delivery of MTN system
CLIN 9 – science payload accommodation

CLIN 4 - 8

*annual requirements met

CLIN 2 – Special Studies

CLIN 3 – Transport to Launch site through Commissioning in Relay Orbit

Fixed Milestones (May Be Vendor Equivalent)

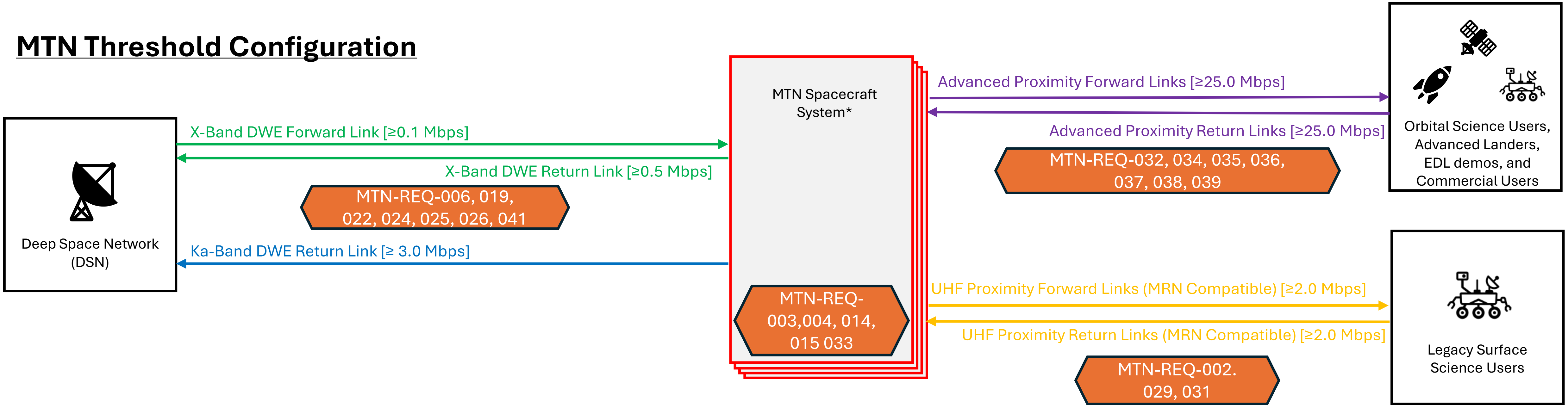
PSR
NLT End of CY 2028

Commissioning
NLT End of CY 2030

- P.L. 119-21 mandates firm, fixed-price contract.
- CLINs separate work packages into logically coherent work units that can be funded independently.
- Offerors are responsible (CLIN 3) for the procurement of the Launch Vehicle, integration, and range coordination.

MTN Threshold Configuration

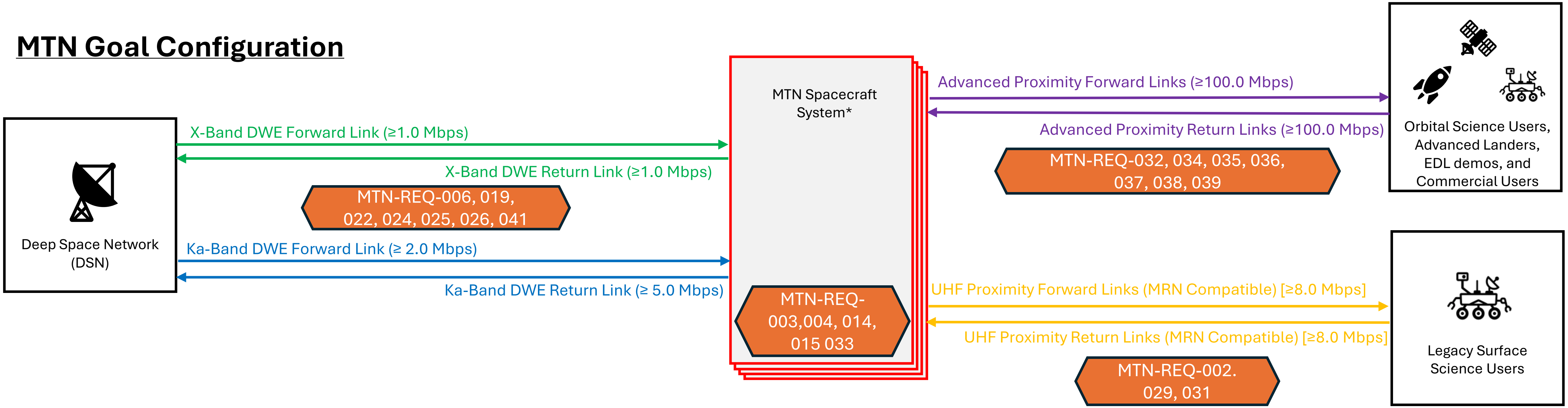
MTN Threshold Configuration



*MTN Spacecraft System may consist of any number of orbital assets depending on contractor architecture

MTN Goal Configuration

MTN Goal Configuration



*MTN Spacecraft System may consist of any number of orbital assets depending on contractor architecture

Insight and Oversight Approach

The MTN draft RFP was formulated with the Starliner failure report in mind

Many of the organizational and technical factors are relevant to a Government-owned, Contractor-operated procurement, although the MTN project is neither human-rated nor service-based. Applicable findings have been incorporated into this draft RFP:

Collaborative Environment:

NASA is committed to establishing and fostering a collaborative environment. Mission success is a mutual objective.

Balancing risk and having the insight necessary to make risk informed decisions:

NASA requires deep technical insight to understand and accept risks as appropriate.

Transparent and Timely Communication:

NASA is committed to embedding Government quality assurance with the vendor and building and maintaining trust.

Leveraging commercial standards and process:

NASA technical standards and processes are one means of assuring mission success and represent years of lessons learned and best practices. NASA also acknowledges that industry has developed and implemented standards that also assure mission success.

The MTN Project intends to collaborate and gather information and data throughout the project life cycle via technical insight as the vendor is performing design, development, testing, and integration activities. NASA understands that some risks are worth taking and balancing risk is a pillar of technical excellence at NASA.

Data Requirements Deliverables (DRDs)

- NASA has defined 54 DRDs
- Intent is that these constitute formal deliveries of information and products that need to produce regardless of their identification as DRDs
- Many (e.g., Risk Management Plan, Configuration Management Plan) may be satisfied by equivalent corporate policy documents
- DRDs are viewed as key deliverables for NASA to maintain appropriate insight

Type	Description
1	NASA reserves a time-limited right to disapprove in writing any versions and interim changes to those versions. The data shall be delivered to NASA for review not less than 20 working days prior to its release for use or implementation. The Contractor shall clearly identify the release target date in the “submitted for review” transmittal. If the Contractor has not been notified of any disapproval prior to the release target date, the data shall be considered approved. To be an acceptable delivery, disapproved data shall be revised to remove causes for the disapproval before its release.
2	NASA reserves a time-limited right to disapprove in writing any versions and interim changes to those versions. The data shall be delivered to NASA for review not less than 10 working days prior to its release for use or implementation. The Contractor shall clearly identify the release target date in the “submitted for review” transmittal. If the Contractor has not been notified of any disapproval prior to the release target date, the data shall be considered approved. To be an acceptable delivery, disapproved data shall be revised to remove causes for the disapproval before its release.
3	These data shall be delivered by the Contractor as required by the contract and do not require NASA approval. NASA may submit comments or questions on these documents for clarification or assistance with understanding of the DRD for the Contractor to consider. However, to be a satisfactory delivery, the data shall satisfy all applicable contractual requirements.

Evaluation Factors

Factors:

- Most advantageous to the Government
- Technical Proposal and Past Performance are significantly more important than Price
- Technical Proposal is significantly more important than Price, and Price is less important than Past Performance

Rationale:

- Schedule is a key component of technical performance
- Combination of schedule and system performance will govern our ability to achieve mission success
- Past performance in relevant work provides strong confidence indicator, particularly on the tight schedule
- Price must fit within available resources

Evaluation Criteria

Written Proposal:

- PWS will be evaluated for feasibility, effectiveness, reasonableness, completeness, and efficiency in meeting the government's objectives.
- Inclusion of all government-defined mandatory work statements from Enclosure 1.

Oral Technical Proposal:

- System's capability to satisfy threshold requirements
- System capabilities exceeding the threshold requirements and approaching or exceeding goal requirements
- Realism of the proposed project schedule
- Management and technical communications
- System interoperability
- System extensibility
- Payments schedule:
    - *Milestones will be evaluated for the technical depth and maturity of the content*
    - *Hardware and results orientated milestones preferred to reporting and reviews*

Past Performance

Typical work elements for spaceflight project

- High-level decomposition of tasks

Relevance:

- Seeks to identify elements of past performance most relevant to MTN
- Table populated by CORs associated with identified contracts

ELEMENT	PERFORMED	DID NOT PERFORM
Project Management		
Risk Management		
Systems Engineering		
Spacecraft Design/Fabrication:		
a. Earth-orbiting mission		
b. Deep space / Lunar / Planetary mission		
c. Space-based Telecommunications Systems		
Spacecraft Integration & Test		
a. Earth-orbiting mission		
b. Deep space / Lunar / Planetary mission		
c. Space-based Telecommunications Systems		
Trajectory Planning, Design, and Navigation:		
a. Earth-orbiting mission		
b. Deep space / Lunar / Planetary mission		
Launch Vehicle Interface Design, Analysis, and Test		
Launch Site Planning, and Shipping		
Launch Vehicle Integration		
Launch Operations		
Post-Launch Vehicle Checkout:		
a. Earth-orbiting mission		
b. Deep space / Lunar / Planetary mission		
c. Space-based Telecommunications Systems		
Mission Operations:		
a. Earth-orbiting mission		
b. Deep space / Lunar / Planetary mission		
c. Space-based Telecommunications Systems		
Spacecraft Safety and Mission Assurance		

Science Payload* Interface & Not to Exceed Envelope

- NASA will choose a science payload.
- MTN-REQ-049 was included in the updated draft RFP, as noted in the cover letter.
- The payload envelope and high-level interface are detailed in the table.
- There is consideration for deploying free-flying CubeSat payloads in Mars orbit.
- A mass simulator will be provided to mitigate schedule risks.
- Schedule: proposals should specify required delivery dates for:
    - science payload interface specification
    - mass simulator
    - payload hardware and related ground support equipment
    - ground system interface specifications and testing.

Criteria	Value	Units
Basic Mass	20	kg
Dimensions	55x55x45	cm x cm x cm
Nominal Power	60	W
Peak Power	60	W
Power Interface Port	28V	
Communication Interface	Wired	
Data Comm Port	RS-422, Ethernet	
Data Rate	25	kbps
Daily Data Volume	200-1000	Mb
Requires Thermal Isolation	No	
Shock Load Limit	GEVS	

*This change is forthcoming in an update to the draft RFP

New Goal: Autonomous Navigation

- P.L. 119-21 language explicitly calls for “autonomous operations, onboard processing, and extended mission duration capabilities.”
- Mars operational environment would benefit tremendously by reducing reliance on Earth-based capabilities
- Reduced loading from Mars on DSN would relieve strain
- Expect future MTN use cases for Mars to drive improved and more frequent navigation updates
- We expect to add a performance goal to augment the functional goal

MTN-REQ-042 *Autonomous Navigation*

Threshold Requirement: N/A

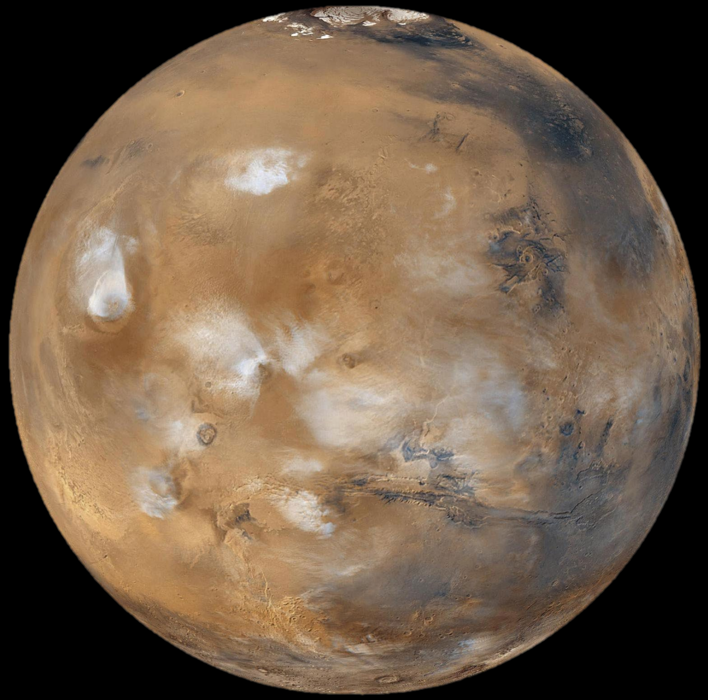
Goal Requirement: MTN shall estimate its navigation state onboard without utilization of Earth-based radiometric tracking.

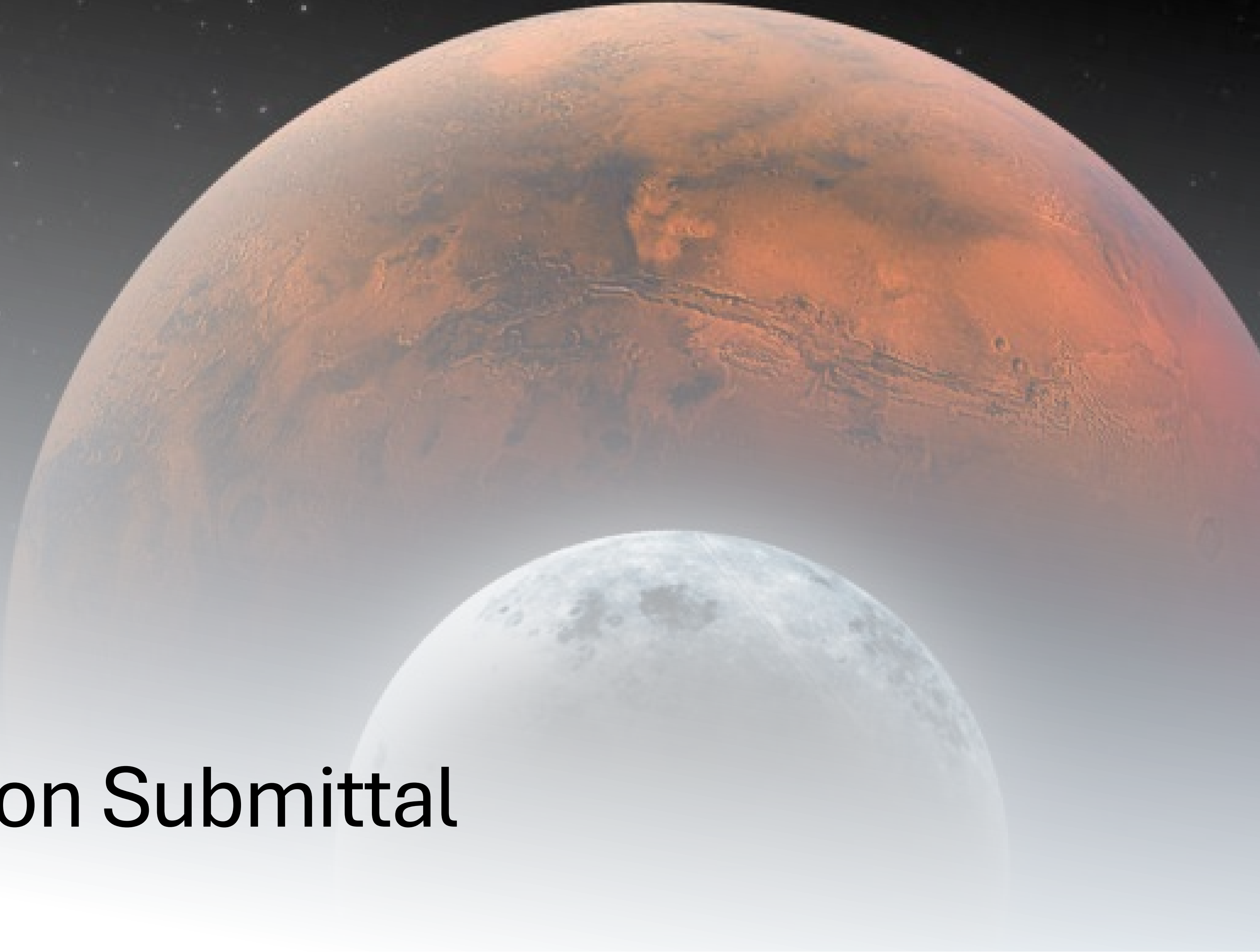
Rationale: MTN is chartered to support autonomous operations and onboard processing. Updating the navigation state onboard is key to enabling more autonomous operations at Mars, which will be critical to systems at Mars operating more independently from Earth.

Verification Method: Test

We Want Your Feeback

- The MTN project highly encourages critical feedback
- To meet the challenging schedule, we are looking for ways to streamline:
  - make processes and documentation serve us rather than being burdens
  - focus on system hardware, software, test -> high-confidence performance / success
- We are committed to maintaining a fair and competitive environment while remaining consistent with P.L. 119-21.
- NASA needs MTN’s capability; help us move fast and deliver on-time success.





Break & Question Submittal

Question & Answer Instructions

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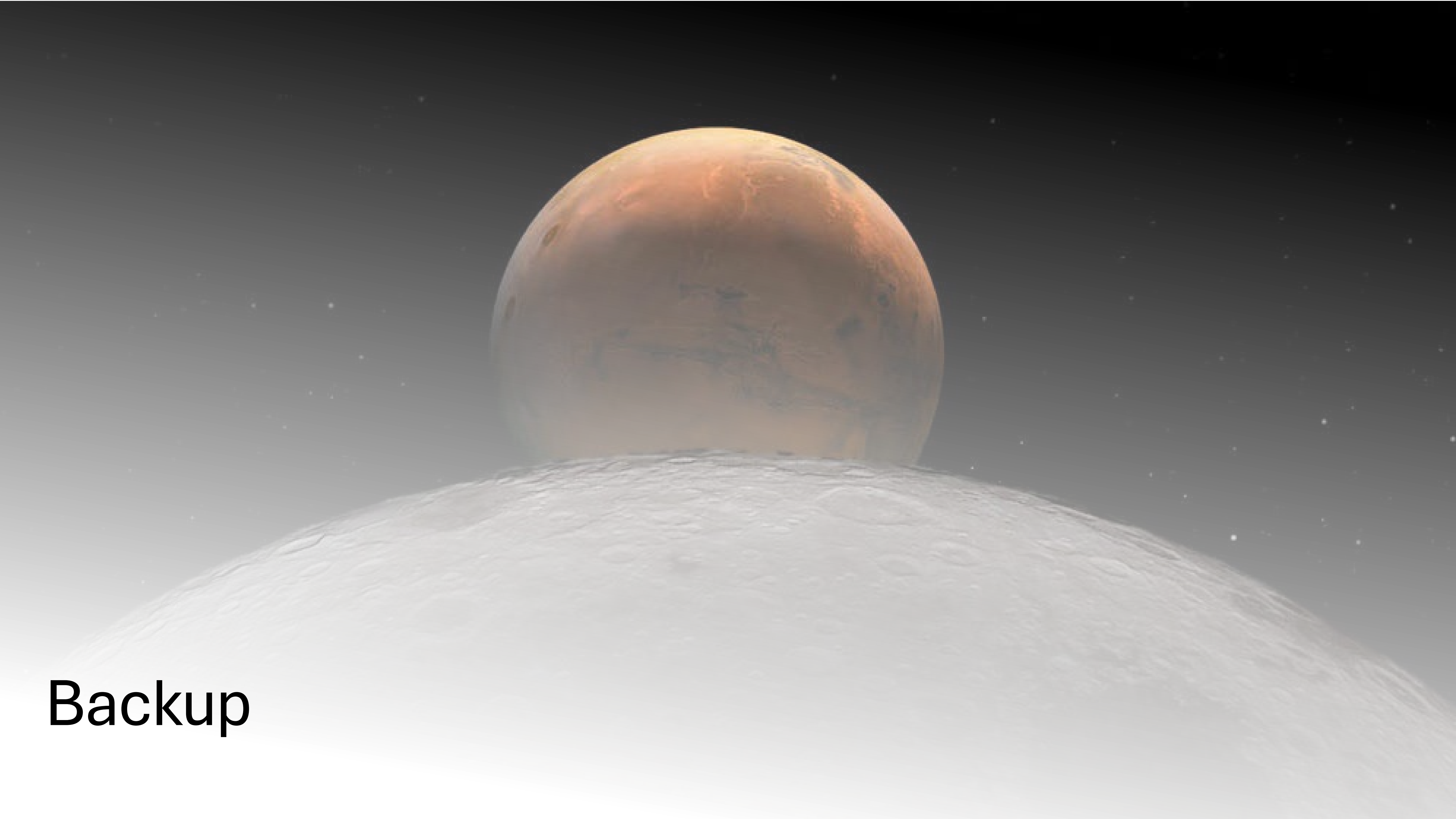

Backup

Cybersecurity Questionnaire (CASQ)

- Seeks to proactively assess industry standard treatment of critical cybersecurity controls for mission operations environments
- Clarify expectations regarding cybersecurity implementations from the beginning of contract
- Leverage standard commercial practices as applied to mission operations environments

References: a. Committee on National Security Systems Policy 12, National Information Assurance Policy for Space Systems used to Support National Security Missions, 28 November 2012 b. National Institute of Standards and Technology Special Publication 800-53, Rev 5, Security and Privacy Controls for Federal Information Systems, December 2020			
Item Number:	**Name**	**Description**	**Vendor's Response:**
Access Control			
AC-2	ACCOUNT MANAGEMENT	Describe how the proposed solution addresses the Information Assurance and Cyber Security concerns stated here: The organization: a. Identifies and selects the following types of information system accounts to support organizational missions/business functions: System Administrator, Network Administrator, Application Administrator, and Database Administrator accounts; b. Assigns account managers for information system accounts; c. Establishes conditions for group and role membership; d. Specifies authorized users of the information system, group and role membership, and access authorizations (i.e., privileges) and other attributes (as required) for each account; e. Requires approvals by System Owner and Authorizing Official for requests to create information system accounts; f. Creates, enables, modifies, disables, and removes information system accounts in accordance with industry standards (e.g. NIST, ISO, FISMA); g. Monitors the use of, information system accounts; h. Notifies account managers: 1. When accounts are no longer required; 2. When users are terminated or transferred; 3. When individual information system usage or need-to-know changes; i. Authorizes access to the information system based on: 1. A valid access authorization; 2. Intended system usage; and 3. Other attributes as required by the organization or associated missions/business functions; j. Reviews accounts for compliance with account management requirements semi-annually; and k. Establishes a process for reissuing shared/group account credentials (if deployed) when individuals are removed from the group.	

Requirements Legend

ID	Title	Threshold Requirement	Rationale	Goal	Type	VM(s)
Requirement ID	Requirement Title	Threshold capability requirement language	Requirement Rationale	Goal capability requirement language	Requirement Type: • Functional (F) • Performance (P)	Anticipated Verification Method (VM): • Analysis (A) • Inspection (I) • Demonstration (D) • Test (T)